



ST BENEDICT'S

SUBJECT	CHEMISTRY EXAMINATION
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mr Hilder
DATE	10 November 2025
MARKS	135
DURATION	2 hours

%

LEARNER NAME: **MEMO**

KEY:	D ☐	E ☐	F ☐	TEACHER:	ZN ☐	KH ☐	FMC ☐
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PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 17 pages and a separate Data Sheet and Periodic Table. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your name and surname in the space provided above AND circle your teacher's name
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. A Blank page is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

Question	1	2	3	4	5	6	Total
Mark							
Marker Initial							
Moderated Mark							
Question Total	27	23	24	23	19	19	135

Comment:

Learner's Signature:

Teacher's Signature:

QUESTION 1**27 MARKS**

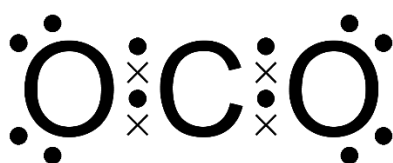
1.1

- 1.1.1 Is a non-polar molecule? (1)
B&E✓
- 1.1.2 Is a non-polar molecule with polar covalent bonds? (1)
B✓
- 1.1.3 Is a polar molecule with polar covalent bonds? (1)
A&F✓
- 1.1.4 Has hydrogen bond forces. (1)
A✓
- 1.1.5 Forms an ionic crystal lattice structure in solid form. (1)
D✓
- 1.1.6 Exists as a solid in a metallic crystal lattice. (1)
C✓
- 1.1.7 Is a solid at room temperature and is soluble in CCl_4 . (1)
E✓

1.2 The structure of solid potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$, is shown in the diagram.

- 1.2.1 Write the oxidation number for chromium in. $\text{Cr}_2\text{O}_7^{2-}$. (2)
6+✓✓
- 1.2.2 Name the type/s of bonds that exist in potassium dichromate. (2)
Polar covalent bond✓ and ionic✓
- 1.2.3 Explain your answer in 1.2.2, with reference to relevant subatomic particles (2)
Electron are shared between Cr and O✓, electrons are transferred from K to Cr_2O_7 ✓

1.3 Draw the Lewis diagram of carbon dioxide and hence explain if there are any mistakes present in the AI image above. (2)



Error - Single bonds✓

Correct sharing of electrons✓

- 1.4 Define the term *electronegativity*. (1)

The measure of the tendency of an atom to attract a bonding pair of electrons ✓.

- 1.5 Use calculations to determine the bond polarity for the C-O bond. (2)

$$\text{END} = 3,5 - 2,5 = 1 \quad \checkmark$$

Polar covalent bond ✓

- 1.6 Name the type of intermolecular forces between carbon dioxide molecules and explain how these intermolecular forces arise. (5)

Both non polar molecule and experience London forces ✓

Random movement of electrons and setting up temporary dipoles ✓

Causing neighbouring atom to disperse. ✓

This results in an induced dipole forming. ✓

The two dipoles are then able to attract. ✓

- 1.7 Carbon dioxide has a melting point of $-78,9^{\circ}\text{C}$, methane has a melting point of -182°C . Fully explain these differences in melting point. (4)

Carbon dioxide and methane are both non polar molecules ✓

Carbon dioxide has higher electron density ✓, Carbon dioxide has stronger London forces ✓

More energy is required to overcome stronger London forces ✓

HI has a higher boiling point.

QUESTION 2

23 MARKS

- 2.1 Define the term *reaction rate* for a chemical reaction. (1)

The change in concentration per unit time of either reactants or products ✓.

- 2.2 Balance the given partially-balanced equation. (2)



- 2.3 Is the above reaction **EXOTHERMIC** ✓ **ENDOTHERMIC**? Circle your answer (1)

- 2.4 Explain your answer in 2.3. (1)

The heat of reaction is less than zero ✓

- 2.5 Morpheus provides Neo with the apparatus shown below. Explain how Neo would use this to measure the rate of the reaction. (2)

Measure the volume/amount of the gas being produced ✓ in the syringe every few seconds ✓

- 2.6 Canva AI originally plotted a concentration vs time graph for this reaction showing all species except for water. Explain why that would be incorrect. (2)

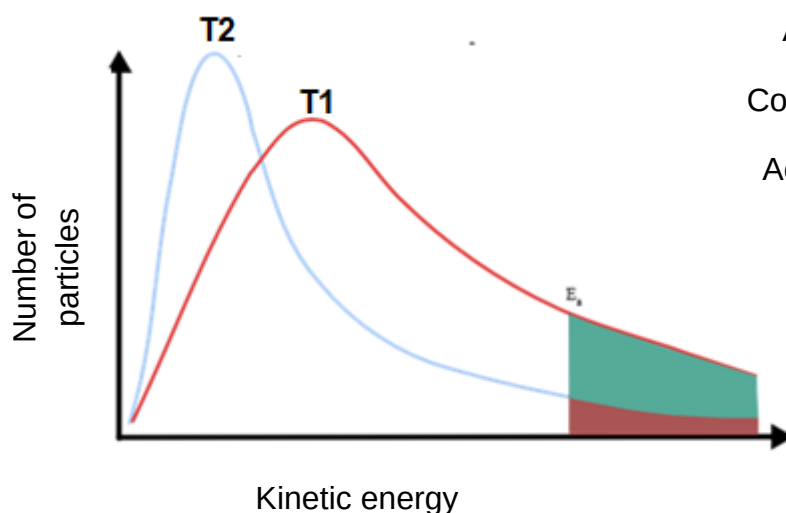
Aluminium is solid. ✓ **OR** Water is a liquid concentration of solids and liquids ✓ and cannot change. ✓

Concentration of the solid cannot be changed. ✓

- 2.7 Calculate the rate of reaction for aluminium. (2)

$$\text{Reaction rate} = 0,5/70 \text{ ✓} = 7,14 \times 10^{-3} \text{ mol.s}^{-1} \text{ ✓}$$

- 2.8 (4)



Axis labelled ✓

Correct shapes ✓✓

Activation energy labelled ✓

2.9 How does a *catalyst* affect the rate of a reaction? (2)

A substance that increases the reaction rate ✓ by choosing the alternative path ✓

2.10 Indicate the effect that the addition of a catalyst has on **the graph** on the previous page.

Ea more to the right ✓ (2)

2.11 Use collision theory to explain whether a change in surface area would affect the rate of this reaction. (4)

Increased surface area exposes more solid reactant particles. ✓

There will be more collisions per unit time ✓

There will be more effective collisions (with correct orientation and kinetic energy greater than activation energy) per unit time ✓

The reaction rate increases ✓

QUESTION 3**24 MARKS**

3.1

3.1.1 Prove that the silver nitrate is the limiting reactant.

(6)

 MgCl_2

$$n = c \cdot V = (0,25)(0,055) = 0,01375 \text{ mol}$$

 AgNO_3

$$n = c \cdot V = (0,15)(0,070) = 0,0105 \text{ mol}$$

 $\text{MgCl}_2 : \text{AgNO}_3$

1 : 2

$$n = \frac{0,0105}{2} = 5,25 \times 10^{-3} \text{ of } \text{MgCl}_2 \text{ required}$$

If AgNO_3 is limiting then $5,25 \times 10^{-3} \text{ mol}$ of MgCl_2 required. 0,01375 available (enough). AgNO_3 limiting.

3.1.2 Hence, calculate the mass of silver chloride that can formed during this reaction.

(3)

 $2\text{AgNO}_3 : 2\text{AgCl(s)}$

2 : 2

$$n = 0,0105 \text{ mol}$$

$$n = \frac{m}{M}$$

$$0,0105 = \frac{m}{143,5}$$

$$m = 1,51 \text{ g answer correct rounding}$$

Unrounded: 1,44 g

- 3.1.3 Calculate the percentage yield of silver chloride if 1,49 g of silver chloride was in fact recovered. (2)

$$\%yield = \frac{actual}{theoretical} = \frac{1,49}{1,51} \times 100 = 98,98\%$$

- 3.2 You are given the following two aqueous solutions:

- 3.2.1 Give a reason why the above solutions are *standard solutions*. (1)

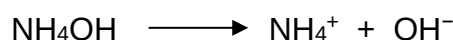
They have known concentrations ✓

- 3.2.2 Define the term *solvent*. (2)

The substance in which another substance is dissolved ✓, forming a solution ✓

- 3.2.3 Identify the solvent used in the above solutions. (1)

The balanced dissociation reaction for ammonium hydroxide is given as:



- 3.2.4 Write balanced chemical equation showing the dissociation of ammonium carbonate. Include phases. (4)



- 3.2.5 Solution A is now mixed with Solution B in a new bigger beaker. Calculate the new concentration of the ammonium ions in the solution. (5)

$$\text{NH}_4\text{OH}: n = c \cdot V = 0,2 \times 0,25 = 0,05 \text{ mol}$$

$$(\text{NH}_4)_2\text{CO}_3: n = c \cdot V = 0,3 \times 0,5 = 0,15 \text{ mol} \times 2 = 0,3 \text{ mol}$$

$$c = \frac{n}{V}$$

$$c = \frac{0,05 + 0,3}{0,25 + 0,50}$$

$$c = 0,47 \text{ mol} \cdot \text{dm}^{-3}$$

QUESTION 4

23 MARKS

- 4.1 Hydrogen gas is bubbled through a blue solution of copper (II) sulphate (CuSO_4). No visible reaction occurs. However, when a piece of zinc metal is placed into the same copper (II) sulphate solution, a reddish-brown solid forms on the zinc surface and the blue colour of the solution fades.
- 4.1.1 Write the oxidation and reduction half-reactions for the reaction that occurs when zinc metal is placed in copper (II) sulphate solution. (4)
- Oxidation: $\text{Zn (s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$
- Reduction: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu (s)}$
- 4.1.2 Use the standard reduction potentials to explain why this reaction is spontaneous, and why the reaction with hydrogen does not occur. (2)
- Zinc is more easily oxidized (has a lower reduction potential) than copper, so zinc reduces Cu^{2+} ions to copper metal spontaneously. OR Zn is a better reducing agent than Cu. Cu^{2+} ions are stronger than Zn^{2+} .
- 4.2 Kwazi now sets up the following electrolytic cell with the aim of coating a key with silver metal, he then dissociated concentrated silver nitrate.
- 4.2.1 Define *reducing agent*. (1)
- A substance that donates electrons (it is oxidised in a redox reaction)
- 4.2.2 Why is the positive electrode made up of silver? (2)
- The silver electrode provides a continuous source of **Ag^+ ions** needed for plating.
- 4.2.3 Show the direction of the silver ion flow in the solution. (1)
- Silver ions (**Ag^+**) move **from the anode (silver electrode) towards the cathode (the key)** in the solution.
- 4.2.4 Write the equation for the half-reaction taking place at the cathode. (2)
- $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag(s)}$
- 4.2.5 A student says that the key is acting as the reducing agent during electroplating of the key with silver. Do you agree with the student's statement? Give a reason for your answer. (2)

No, the student is not correct. (No mark if they only stated this)

The key is the cathode, and at the cathode reduction takes place. OR The key is not a reducing agent.

In this circuit, the power source supplies the electrons, not the key to reduce the silver ions to silver metal.

Therefore, the key does not act as the reducing agent, it simply receives electrons and gets coated with silver.

- 4.3 A CONCENTRATED solution of copper (II) bromide is electrolysed using carbon (graphite) electrodes. The diagram shows a representation of this cell.

Bubbles are observed at the cathode.

- 4.3.1 Which electrode is the cathode? State only **P** or **Q**. Circle your answer. (1)

- 4.3.2 Write the equation for the half-reaction taking place at the anode. (2)



- 4.3.3 Define *oxidation number*. (1)

Oxidation as the loss or gain of electrons

- 4.3.4 Use oxidation number to explain which substance is oxidised and which is reduced. (3)

Cu^{2+} changes from +2 to 0, and get reduced.

Br^- changes from -1 to 0, and get oxidized.

- 4.3.5 Write the balanced net ionic equation for this reaction. (2)



QUESTION 5

19 MARKS

Consider the two hydrocarbons, compounds **X** and **Y**, shown below.

- 5.1 Define *homologous series*. (2)

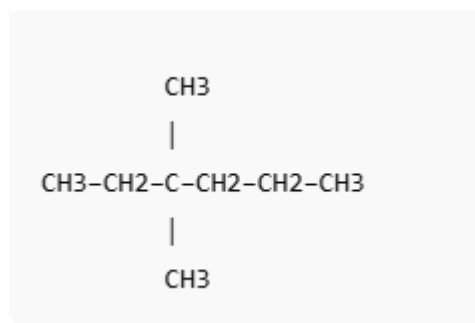
A series of similar compounds that have the same functional group and have the same general formula ✓, in which each member differs from the previous one by a single CH_2 unit ✓.

- 5.2 Write down the general formula for the homologous series to which compound **X** belongs. (2)

$\text{C}_n\text{H}_{2n+2}$ ✓ ✓

- 5.3 Classify compound **X** as **SATURATED** ✓ ✓ or **UNSATURATED**. (2)

- 5.4 Draw the structural formula for the POSITIONAL ISOMER of compound **X**. (2)



Correct parent chain ✓

Correct branches ✓

- 5.5 Write the condensed formula for a chain isomer of compound **X**. (2)

$\text{CH}_3\text{CH}_2\text{C}(\text{C}_2\text{H}_5)(\text{CH}_3)\text{CH}_2\text{CH}_3$

Must have 6 carbons ✓

Correct branches ✓

- 5.6 Describe the functional group of compound **Y**. (1)

Double carbon-carbon bond ✓

- 5.7 Write the IUPAC name for compound **Y**. (3)

3-ethylhexa-1,3-diene (No space between ethyl and hexa and diene)

- 5.8 Which compound is expected to have the **higher boiling point**? (1)

Compound **Y** has the higher boiling point ✓

5.9 Explain your answer in 5.8. (4)

Both X and Y are non-polar hydrocarbons, so the main intermolecular force is London (dispersion) forces. ✓

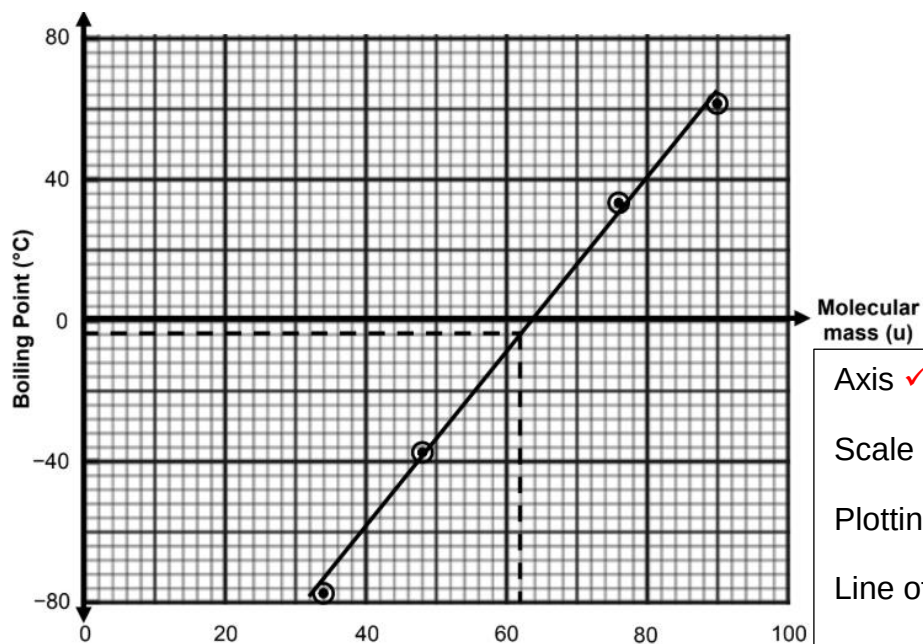
Compound Y is more linear/less branched than X, so it can pack closer and has a larger effective surface area for intermolecular contact. ✓

Greater surface area → stronger dispersion forces → more energy needed to separate molecules → higher boiling point. ✓

Compound X is more highly branched, giving a more compact shape with smaller surface contact between molecules, so it has weaker dispersion forces and a lower boiling point. ✓

QUESTION 6**19 MARKS**

- 6.1 Identify the predominant intermolecular force in CH_3F . (2)
Dipole-dipole interactions ✓✓
- 6.2 Is fluorobutane a GAS or a LIQUID at room temperature (25°C)? (1)
Liquid ✓
- 6.3 Plot a graph of boiling point versus molecular mass for the given data on the axes provided on below. Draw a straight line of best fit. (5)



Axis ✓

Scale ✓

Plotting 2 points ✓✓

Line of bestfit ✓

6.4

 $-4 \pm 2\text{ }^{\circ}\text{C}$ ✓✓ (2)

6.5

6.5.1 Can Jerome use this graph (from Question 6.3) to accurately determine the boiling point of chloromethane? YES/ **NO** ✓. Circle the correct answer. (1)

6.5.2 Briefly explain your answer. (2)

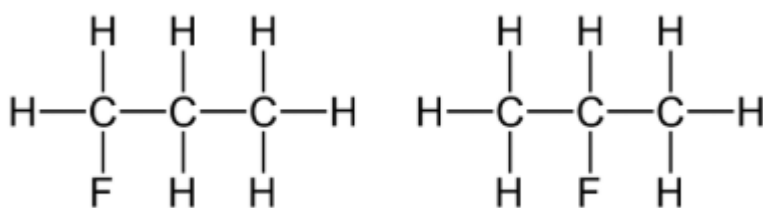
This is because the graph only has data for fluoroalkanes ✓, not chloroalkanes (a different haloalkane was used) ✓.

6.6

6.6.1 Define structural *isomers*. (2)

Compounds having the same molecular formula but different structural formulae ✓✓.

6.6.2 Draw the structural formulae for TWO structural isomers of $\text{C}_3\text{H}_7\text{F}$. (3)



Correct parent chain ✓

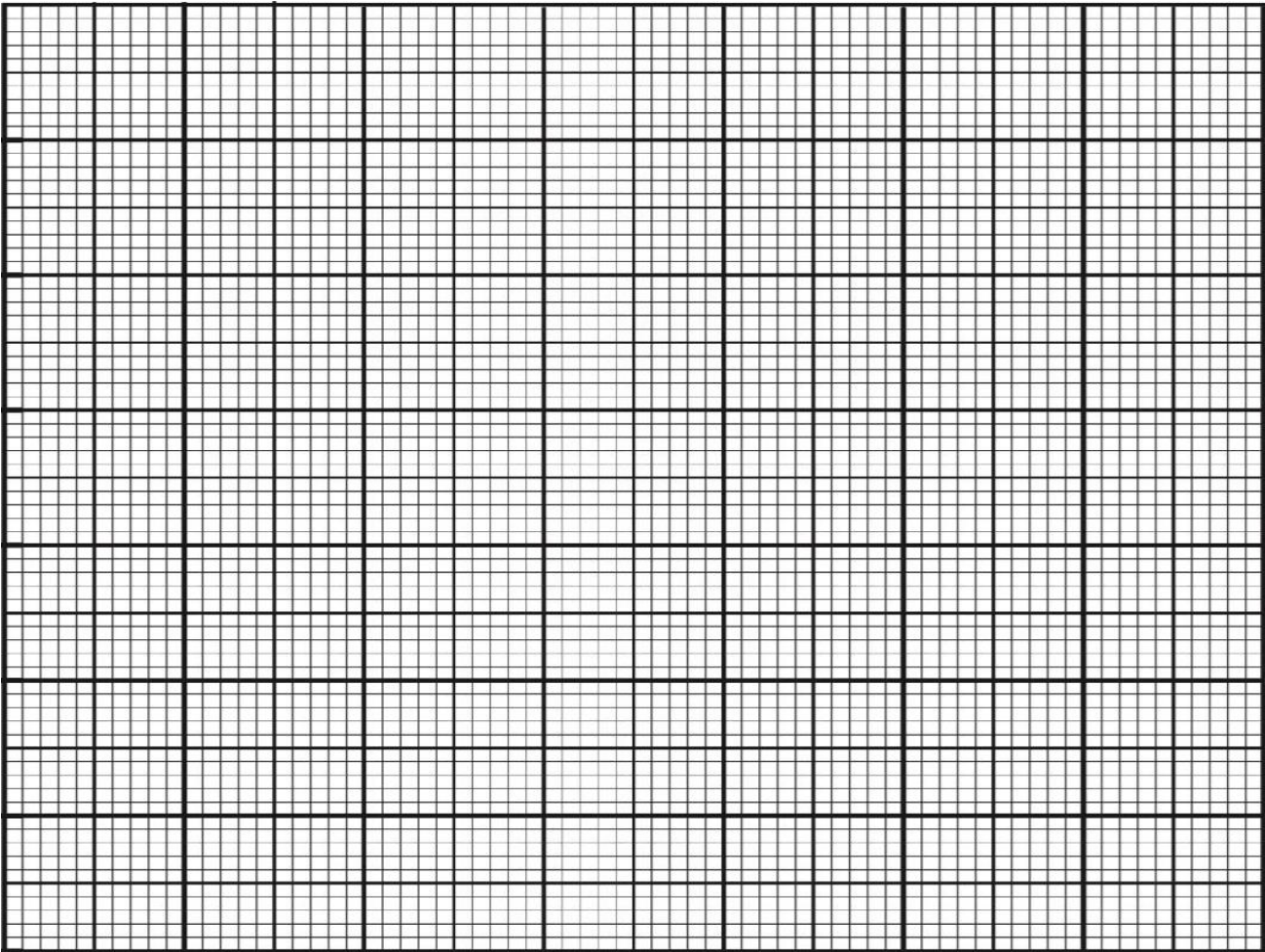
TWO structural isomers ✓✓

6.6.1 Identify the TYPE of structural isomers you have drawn. (1)

Positional isomers ✓

END OF ASSESSMENT

ADDITIONAL GRAPH PAPER



ADDITIONAL SPACE

[illegible]